

BAAVANSH REDDY GUNDLAPALLI

BACKEND AND MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Backend and AI systems engineer (B.S. Computer Science, Rutgers, May 2026) with four years of engineering internships, externships, and a current role building Python REST services that integrate LLM and machine-learning components. Personal projects cover the ML lifecycle end to end: Feast features, MLflow tracking and model registry with promotion and rollback, FastAPI serving, Prometheus and Evidently monitoring, SHAP and Fairlearn evaluation, and PyTorch, TensorFlow/Keras, DDP, and ONNX Runtime benchmarking, each built, tested, and measured locally.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch (DDP, AMP), TensorFlow/Keras, scikit-learn, NumPy, ONNX Runtime, sentence-transformers

MLOps & Serving: MLflow (experiment tracking, model registry), Feast feature store, FastAPI model serving, Docker, Docker Compose, GitHub Actions CI/CD

Evaluation & Monitoring: SHAP, Fairlearn, Evidently, Prometheus, model cards, data drift and training-serving skew checks, Locust

Data & Cloud: PostgreSQL, MySQL, SQLite, Supabase, AWS, MinIO

Testing & Quality: pytest, JUnit, ruff, mypy, bandit

Programming: Python, Java (Spring Boot), SQL, C++, TypeScript (React, Next.js), Linux/Bash

EXPERIENCE

TAIRC (The AI Research Center)

Jun 2026 to Present

Backend and AI Systems Engineer

- Build and maintain Python backend services and REST APIs, designing relational schemas and data-access layers that support AI-enabled research and production applications.
- Integrate LLM and machine-learning components into backend services, covering model invocation, prompt orchestration, retrieval, and output evaluation.
- Convert research prototypes into deployable, tested endpoints, owning features from requirements analysis through API design, implementation, testing, and deployment.
- Work with research and engineering staff to translate technical requirements into production systems, applying software engineering fundamentals across the full development lifecycle.

LIVELY (Jillcyn Enterprises, LLC) *via Rutgers MBS Externship Exchange*

Jan 2026 to May 2026

Full-Stack Engineering Extern

- Co-built LIVEY, a full-stack event-discovery platform, developing React search and filtering components and engineering REST API routes across the front and back end.
- Integrated Supabase (PostgreSQL) for real-time data storage, user authentication, and event management across the platform.

Little Red Riding Hood Inc. *via Rutgers MBS Externship Exchange*

Sep 2025 to Dec 2025

Software Engineering Extern

- Engineered secure authentication for a safety-focused rideshare platform, implementing role-based login and credential management for separate rider and driver accounts.
- Built driver session tracking and real-time geolocation handling on cloud-deployed (AWS) services, supporting dispatch coordination and passenger-safety workflows.

KAVACH (VIANSEC Solutions Pvt. Ltd.)

Sep 2023 to Aug 2025

Backend and IoT Engineering Intern

Aug 2024 to Aug 2025

- Collaborated with the engineering team to integrate IoT telemetry and AI-generated detections into incident and alert workflows.
- Validated device-data flows, monitored system behavior, documented integration issues, and supported troubleshooting.

Software Engineering Intern

Sep 2023 to Aug 2024

- Contributed to dashboard functionality for user access, site operations, incident tracking, and alert management.
- Performed software testing, documented defects, and supported Docker-based deployments and application debugging.

- Contributed to development of a Java and Spring Boot work-order and vendor-tracking application backed by MySQL.
- Tested and documented application features while maintaining database records and operational reports.

SELECTED ML PROJECTS (PERSONAL AND ACADEMIC)

MLOps Deployment Lab | PyTorch, Feast, MLflow, FastAPI, Docker, Prometheus

github.com/BaavanshReddy/mlops-deployment-lab

- Built an end-to-end local ML pipeline on the UCI Adult dataset: contract-validated ingestion with leakage checks, Feast offline and online features with an automated training-serving parity test, and scikit-learn and PyTorch (ONNX) models tracked in MLflow with a registry promotion policy and rollback.
- Served models through FastAPI with async micro-batching (1.9x the unbatched throughput at 64 concurrent users in local Locust tests) on Docker Compose with Prometheus alert rules and Evidently drift alerts; wrote Kubernetes blue-green manifests validated against a Kubernetes API server; 93 automated tests.

Responsible AI Evaluation (MLOps Lab extension) | SHAP, Fairlearn, Evidently, MLflow

- Evaluated each registered model version with permutation SHAP (additivity error below $1e-14$), Fairlearn subgroup diagnostics with 300-resample bootstrap intervals and a minimum-size safeguard that excluded 9 undersized groups from comparisons, and threshold sweeps across sex, race, age, and native-country slices.
- Generated model cards and SHA-256-hashed release-evidence records tied to the registry version, dataset version, and Git commit, with a lint that blocks fairness or approval claims; a controlled input shift raised Evidently alerts and moved the sex equalized-odds gap from 0.167 to 0.301.

Neural Network Training Benchmark | NumPy, PyTorch, Keras, ONNX

github.com/BaavanshReddy/neural-net-from-scratch

- Rebuilt a from-scratch NumPy network (manual backpropagation) in PyTorch and Keras on shared data, weights, and batch order; in float64 all three, plus gradient accumulation and 2-process CPU distributed data-parallel training (DDP, gloo), agree within $1e-15$, reaching 90.2% (digits) and 90.9% (faces) test accuracy over 3 seeds.
- Added DataLoader tuning, CPU bfloat16 mixed precision, bitwise-identical checkpoint resume, TensorFlow MirroredStrategy, and ONNX export (max probability gap $1.0e-6$ versus PyTorch), plus training, batch, and concurrent inference benchmarks; ONNX Runtime halved single-record p50 latency versus PyTorch (15 vs 32 microseconds).

LLM FactCheck | Python, BM25, RAG, TriviaQA, YAMLL

github.com/BaavanshReddy/LLM_factcheck

- Co-authored a two-person course research paper and evaluation pipeline comparing direct LLM answering, BM25 retrieval, and BM25-grounded RAG on a 100-question TriviaQA pilot; RAG raised normalized exact match from 0.66 to 0.69 and token F1 from 0.784 to 0.803.
- Scored answers with exact match, token F1, a lexical groundedness proxy, and an unsupported-answer proxy, backed by an eleven-category manual error review.

AgentMemry | Python, SQLite, sentence-transformers, NumPy, pytest

pypi.org/project/agentmemry

- Published a Python package on PyPI for persistent agent memory using local embeddings, SQLite storage, cosine-similarity retrieval, and Maximal Marginal Relevance ranking.
- Designed per-agent namespaces and retrieval APIs, covered by 15 pytest tests for search, metadata, updates, deletion, and isolation.

EDUCATION

Rutgers University, New Brunswick, NJ

May 2026

Bachelor of Science in Computer Science, *Magna Cum Laude* | GPA 3.76 / 4.0**Dean's List:** Fall 2024, Spring 2025, Fall 2025, Spring 2026**Relevant Coursework:** Artificial Intelligence, Databases, Operating Systems, Software Engineering, Computer Architecture, Data Structures & Algorithms, Compilers

LEADERSHIP & ACTIVITIES

- **Community Service Officer**, Rutgers University Police Department (*Jan 2025 to May 2026*). Supported incident response and crowd management for large-scale university events.
- **Leadership Roles**, Alpha Phi Delta Fraternity (*May 2024 to May 2026*). Held three elected positions (Risk Management Chair, Head of Judiciary Board, Co-Brotherhood Chair); led safety compliance, conduct governance, and member engagement.
- **Administration Secretary**, ENIGMA Technical Society (*Aug 2023 to May 2024*). Organized technical workshops and streamlined communication across society events.

Languages: English, Hindi, Telugu